Experiment No. 5
C++ CONDITIONAL STATEMENTS

I. OBJECTIVES

a. To implement the different conditional statements in C++ programs.
b. To be able to identify the application and limitations of the different conditional statement.

II. LABORATORY EXERCISES

Encode the following program and compile them. Illustrate the output.
Note: if getch(); will not work on other C++ ide compiler replace it with _getch();
For www.onlinegdb.com editor and cxxdroid (mobile app) you may remove it.

Program no.1 (save as ifex1.cpp)

```
#include<iostream>
#include<conio.h>
// this program will display a message if the entered number is 10
using namespace std;
int main ()
{

int a;
cout<<"\n"<<"Enter a number : " ;
cin>>a;
if (a==10)
      cout<<a<<" is equal to 10. " ;
_getch();
return 0;
}
```

Program No. 2 (save as ifex2.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
// this program will test if the number is positive or negative

int main ()
{
int a;
cout<<"\n"<<"Enter a number: ";
cin>>a;
if(a<0)
      cout<<"The number is negative";
else
      cout<<"The number is positive or zero: ";
_getch();
return 0;
}
```

Program No 3. (save as ifex3.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
// this program will test if the number is positive, negative or zero

int main ()
{
int a;
cout<<"\n"<< "Enter a number : ";
cin>>a;
if(a<0)
cout<<"The number is negative.";
else if(a>0)
      cout<<"The number is positive.";
else
      cout<<"The number is zero";

_getch();
return 0;
}
```

Program No. 4(save as swex1.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
int main ()
{
int code;
cout<<"Enter a code: " ;
cin>>code;
switch (code) {
      case 1: cout<<"\nFirst Year" ;
            break;
      case 2: cout<<"\nSecond Year" ;
            break;
      case 3: cout<<"\nThird Year" ;
            break;
      case 4: cout<<"\nFourth Year" ;
            break;
      case 5: cout<<"\nFifth Year" ;
            break;
      default:
            cout<<"\nNot in the choices" ;
      }
_getch();
return 0;
}
```

Program No. 5(save as prog5.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
int main ()
{
char LtrGrade;
cout <<"Enter a letter grade: " ;
cin>> LtrGrade;
switch (LtrGrade) {
case 'a':
case 'A': cout<<"\nExcellent" ;
      break;
case 'b':
case 'B': cout<<"\nSuperior" ;
break;
      case 'c':
      case 'C': cout<<"\nAverage" ;
            break;
      case 'd':
      case 'D': cout<<"\nPoor" ;
            break;
      case 'f':
      case 'F': cout<<"\nTry Again" ;
            break;
      default:
          cout<<"\nNo match was found for the ENTRY "<<LtrGrade<<endl;
            }
_getch();
return 0;
}
```

III. Summary of Program Outputs

Direction: Demonstrate the corresponding output for each of the given programs. Give your observation and analysis for each of the problems.

1.

```
C:\Users\Admin\OneDrive - CPC\Proj
Enter a number : 10
10 is equal to 10.
```

This program is a type of conditional statement in C++ that uses the basic If statement to determine if the entered number is equal to 10. If so, it will display the text "10 is equal to 10", but if the entered number is not equal to 10, then it will show nothing.

2.

```
C:\Users\Admin\OneDrive - CPC\Proje
Enter a number: 5
The number is positive or zero:
```

This program is a type of conditional statement in C++ that uses the If/Else Statement to determine whether the inputted number is positive or zero or negative. If the inputted number is positive, it will display the text " The number is positive or zero", but if the number is negative, it will show "The number is negative".

3.

```
C:\Users\Admin\OneDri
Enter a number : 0
The number is zero
```

This program is an advanced version of the program No.2. This program uses a ladderized if/else statement to determine whether the inputted number is positive, negative, or zero. If the inputted number is positive, it will display the text " The number is positive". If the number is negative, it will show "The number is negative," and if the number is equal to zero, it will show "The number is zero."

4.

```
C:\Users\Admin\C
Enter a code: 1
First Year
```

This program is an example of a conditional statement in C++. It uses the switch statement method to determine if the entered code is equal to what year from1–5. If the code is equal to "1", it will display the text "First Year", if the code is equal to "2", it will display the text "Second Year", if the code is equal to "3", it will display the text "Third Year", if the code is equal to "4", it will display the text "Fourth Year", if the code is equal to "Fifth Year", and if the code is not equal to 1–5, then the program will display the text "not in the choices".

5.

```
C:\Users\Admin\OneDrive - CP
Enter a letter grade: f

Try Again
```

This program is an example of a conditional statement in C++. It uses the switch statement method to determine the corresponding value of the inputted letter grade in A, B, C, D, and F. If the letter grade is equal to "A/a" it will display the text "Excellent", if the letter grade is equal to "B/b" it will display the text "Superior", if the letter grade is equal to "C/c" it will display the text "Average", if the letter grade is equal to "D/d" it will display the text "poor", and if the letter grade is equal to "F/f" it will display the text "Try Again", but if the text is not in the list, then it will show the text "No match was found for the ENTRY".

IV. Supplementary Problems

1. Write a "scissor-rock-paper" game, commonly known as "jack n poy" in the Philippines. Two players will input the first equivalent letter. For example, s for scissor, r for rock, and p for paper. Then display the winner and the basis of winning such as: "paper covers rock", "rock breaks scissors", and "scissors cut paper", or "nobody wins". Capital or small letters should be accepted by your program.

```
#include <iostream>
#include <conio.h>
using namespace std;


int main()
{
   char P1, P2;

        cout << "Please use 'R/r' for Rock, 'S/s' For Scissor, and 'P/p' for Paper then press ENTER\n";
        cout << "Enter Player 1: ";
        cin >> P1;
        cout << "Enter player 2: ";
        cin >> P2;

        if ((P1 == 'R' || P1 == 'r') && (P2 == 'S' || P2 == 's'))
                cout << "Rock breaks Scissors so Player 1 WIN!! ";
        else if ((P1 == 'S' || P1 == 's') && (P2 == 'R' || P2 == 'r'))
                cout << "Rock breaks Scissors so Player 2 WIN!! ";
        else if ((P1 == 'S' || P1 == 's') && (P2 == 'P' || P2 == 'p'))
                cout << "Scissor cut Paper so Player 1 WIN!! ";
        else if ((P1 == 'P' || P1 == 'p') && (P2 == 'S' || P2 == 's'))
                cout << "Scissor cut Paper so Player 2 WIN!! ";
        else if ((P1 == 'P' || P1 == 'p') && (P2 == 'R' || P2 == 'r'))
                cout << "Paper covers Rock so Payer 1 WIN!! ";
        else if ((P1 == 'R' || P1 == 'r') && (P2 == 'P' || P2 == 'p'))
                cout << "Paper covers Rock so Payer 2 WIN!! ";
        else
                cout << "Nobody Wins";

        _getch();
        return 0;
}
```

OUTPUT:

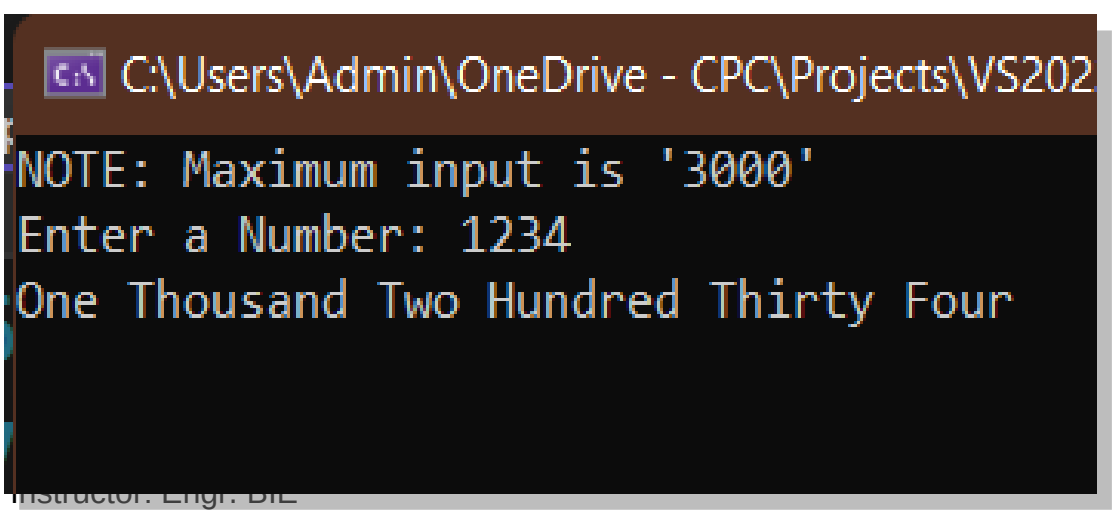
```
C:\Users\Admin\OneDrive - CPC\Projects\VS2022 Projects\Jack N Poy 2.0\x64\Debug\Jack N Poy 2.0.exe
Please use 'R/r' for Rock, 'S/s' For Scissor, and 'P/p' for Paper then press ENTER
Enter Player 1: R
Enter player 2: s
Rock breaks Scissors so Player 1 WIN!!
```

2. Write a program that accepts an ordinary number and outputs its equivalent in words. For example, the inputted number is 1380 the output will be "one thousand three hundred eighty". Take note that the maximum input number is 3000.

```
#include <iostream>
#include <conio.h>
using namespace std;

int main()
{
        int num;
        cout << "NOTE: Maximum input is '3000' \n";
        cout << "Enter a Number: ";
        cin >> num;

        int thou = num / 1000;
        int hun = (num - thou * 1000) / 100;
        int ty = (num - (thou * 1000) - (hun * 100)) / 10;
    int teen = num - (thou * 1000) - (hun * 100);
        int one = (num - (thou * 1000) - (hun * 100) - (ty * 10));


        switch (thou) {
        case 1:cout << "One Thousand "; break;
        case 2:cout << "Two Thousand "; break;
        case 3:cout << "Three Thousand "; break;
        }

        switch (hun) {
        case 1: cout << "One Hundred "; break;
        case 2: cout << "Two Hundred "; break;
        case 3: cout << "Three Hundred "; break;
        case 4: cout << "Four Hundred "; break;
        case 5: cout << "Five Hundred "; break;
        case 6: cout << "Six Hundred "; break;
        case 7: cout << "Seven Hundred "; break;
        case 8: cout << "Eight Hundred "; break;
        case 9: cout << "Nine Hundred "; break;
        }
        switch (ty) {
        case 2: cout << "Twenty "; break;
        case 3: cout << "Thirty "; break;
        case 4: cout << "Forty "; break;
        case 5: cout << "Fifty "; break;
        case 6: cout << "Sixty "; break;
        case 7: cout << "Seventy "; break;
```

```
        case 8: cout << "Eighty "; break;
        case 9: cout << "Ninety "; break;
        }

        switch (teen) {
        case 11:cout << "Eleven"; break;
        case 12:cout << "Twelve"; break;
        case 13:cout << "Thirteen"; break;
        case 14:cout << "Fourteen"; break;
        case 15:cout << "Fiveteen"; break;
        case 16:cout << "Sixteen"; break;
        case 17:cout << "Seventeen"; break;
        case 18:cout << "Eightteen"; break;
        case 19:cout << "Nineteen"; break;
        }

        switch (one) {
        case 1:cout << "One"; break;
        case 2:cout << "Two"; break;
        case 3:cout << "Three"; break;
        case 4:cout << "Four"; break;
        case 5:cout << "Five"; break;
        case 6:cout << "Six"; break;
        case 7:cout << "Seven"; break;
        case 8:cout << "Eight"; break;
        case 9:cout << "Nine"; break;
        }
        if (num > 3000) {
                cout << "THE MAXIMUM INPUT IS '3000'";
        }
        if (num == 0) {
                cout << "Zero";
        }
        _getch();
        return 0;
}
```

OUTPUT:



V. Conclusion

I learned that conditional statements in C++ are used to make choices dependent on a specific condition. If the condition evaluates to True, one set of statements is carried out; otherwise, another set of statements is conducted. In a conditional statement, you can utilize the if statement, the if/else statement, the ladderized or nested if/else statement, or the case switch statement. The if statement chooses and executes a statement depending on a specified condition. If the condition evaluates to True, a specified set of statements are performed. If the condition evaluates to False, however, the supplied sequence of statements is bypassed, and control is passed to the statement that follows the if statement. Depending on the condition's result, the if–else statement causes one of two alternative sentences to execute. The ladderized or nested if/else statement consists of many if-else statements. The switch statement takes a set of possible subsets of statements. The switch statement evaluates the value of an expression within a sequence and compares it to a list of integer or character constants. character constants.